Lab Exercise 19- ClusterIP in Service in Kubernetes

A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas:
  selector:
    matchLabels:
      app: nginx   # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx  # MUST match service selector
    spec:
      containers:
      - name: nginx
        image: nginx
        ports:
        - containerPort: 80
```

Save file.

```
kubectl apply -f nginx-deployment.yaml
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f nginx-deployment.yaml

deployment.apps/nginx-app created
mohdanas@Mohds-MacBook-Air Desktop %
```

Verify:

```
kubectl get pods --show-labels
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get pods --show-labels
NAME                          READY   STATUS    RESTARTS   AGE     LABELS
nginx-app-75fdcbbc74-66x2h    1/1     Running   0          18s     app=nginx,pod-template-hash=75fdcbbc
74
nginx-app-75fdcbbc74-gvgds    1/1     Running   0          18s     app=nginx,pod-template-hash=75fdcbbc
74
test-vol1                     1/1     Running   0          9m13s   <none>
test-vol2                     1/1     Running   0          4m39s   <none>
mohdanas@Mohds-MacBook-Air Desktop %
```

You should see:

```
app=nginx
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```
apiVersion: v1
kind: Service
metadata:
```

```
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx   # MUST match pod label
  ports:
  - port: 80
    targetPort: 80
```

Apply:

```
kubectl apply -f nginx-service.yaml
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl apply -f nginx-service.yaml

service/nginx-service created
```

Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get svc

NAME            TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
nginx-service   ClusterIP   10.96.173.192   <none>        80/TCP    15s
mohdanas@Mohds-MacBook-Air Desktop %
```

You should see:

```
TYPE       CLUSTER-IP
ClusterIP  10.x.x.x
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME            ENDPOINTS                       AGE
nginx-service   10.244.0.64:80,10.244.0.65:80   30s
mohdanas@Mohds-MacBook-Air Desktop %
```

Now you should see something like:

```
10.244.x.x:80, 10.244.x.x:80
```

Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

Inside pod:

```
wget -qO- http://nginx-service
```

You should see:

```
Welcome to nginx!
```

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl run test-pod --image=busybox --rm -it -- sh
If you don't see a command prompt, try pressing enter.
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ #
```

Exit:

```
exit
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

It will NOT work.

```
mohdanas@Mohds-MacBook-Air Desktop % kubectl port-forward service/nginx-service 8080:80
Forwarding from 127.0.0.1:8080 -> 80
Forwarding from [::1]:8080 -> 80
Handling connection for 8080
```

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

That proves:

ClusterIP works only inside cluster.